ADVERSARIAL ATTACKS ON LARGE LANGUAGE MODELS

UGRC EE4902

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ABSTRACT

Hallucinations in large language models (LLMs) refer to responses that are factually incorrect, fabricated, or unsupported by existing data. While much of the prior research has focused on defining metrics to detect and quantify hallucinations, recent interpretability efforts attempt to uncover the internal circuits within Transformer architectures that give rise to such phenomena. In this work, we adopt a behavioral perspective, analyzing the response of white-box LLMs to adversarial perturbations applied directly in logit space. Our study examines the robustness of these models to injected errors, the mechanisms of error propagation, why LLMs hallucinate, understanding if top-1 selection (greedy) retrieves the information obtained from nuclear sampling and the extent to which internal model dynamics enable error correction. Drawing a parallel with classical communication systems, we conceptualize the LLM as a noisy channel, where the transformation from internal representations to output text is susceptible to perturbations. This analogy allows us to frame hallucinations as a form of information corruption and to analyze the model's internal pathways through the lens of error detection and correction, akin to coding theory in digital communication.

Keywords Adversarial Attacks · Large Language Models · Hallucinations

1 Introduction

Large Language Models (LLMs) represent the current state-of-the-art in text generation and various downstream tasks. As their generative abilities continue to scale with model size, issues such as factual inaccuracies and biased or insensitive responses (e.g., gender bias) remain active areas of investigation. A model is said to hallucinate when it produces outputs that contradict established facts or are unsupported by existing data. Recent research efforts aim to identify the internal layers responsible for such behavior. Many of these studies use misleading one-shot prompts containing false information, a technique we refer to as an **embedding space attack**.

In contrast, our work explores the effects of directly injecting errors into the model's logit space. We examine how these perturbations propagate during the forward pass, leveraging the autoregressive structure of Transformer-based models. Inspired by **Communication Systems**, we conceptualize an LLM as a noisy communication channel, with the prompt serving as the input signal. Our experiments assess the model's robustness to such perturbations by evaluating the generated text both quantitatively and qualitatively.

Through these **logit space attacks**, we find that LLMs exhibit a degree of error correction, often maintaining the linguistic coherence and grammatical structure of the generated output despite the injected noise.

2 Related Work

Adversarial robustness and error propagation in autoregressive language models have recently attracted considerable attention. A key challenge in sequence generation is the so-called exposure bias, where models are trained on ground-truth contexts but generate tokens conditioned on previously predicted outputs during inference, causing errors to accumulate over time Bengio et al. [2015]. Methods such as scheduled sampling Bengio et al. [2015] and imitation

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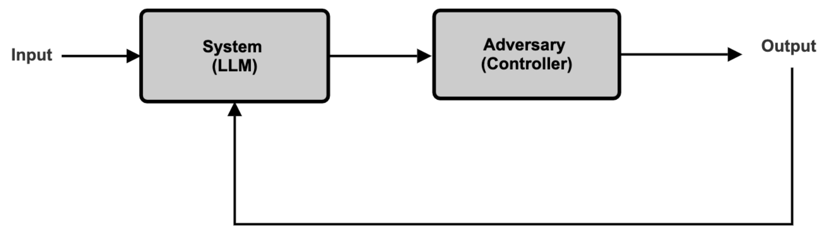

Figure 1: Logit Space Attack

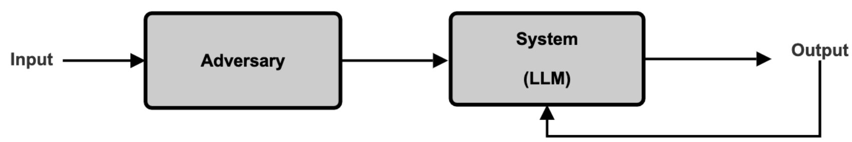

Figure 2: Input Space Attack

learning algorithms like DAGGER Ross et al. [2011] have been proposed to alleviate this discrepancy by gradually introducing model-generated tokens during training, highlighting the sensitivity of language models to perturbations in their context. Complementing this, Khandelwal et al. Khandelwal et al. [2019] empirically demonstrated that minor perturbations early in the generated sequence can lead to drastically different completions, underscoring the importance of understanding how errors propagate in the autoregressive generation process.

Recent work has shifted towards analyzing the internal representations of language models, particularly focusing on the logit space—the vector of unnormalized probabilities output before decoding. Liu et al. Liu et al. [2022] introduced Direct Logit Attribution (DLA), which decomposes logits to reveal contributions from specific neurons or residual streams, providing insights into the internal decision-making process of Transformers. Several studies have leveraged logit-level perturbations to mount adversarial attacks or steer model outputs Wallace et al. [2019], Zhang et al. [2023], though these often focus on single-step manipulations rather than multi-step propagation effects. Carlini et al. Carlini et al. [2023] examined adversarial examples that can cause targeted output changes, pointing to the fragility of these models under crafted input perturbations. However, comprehensive studies of how injected noise or adversarial perturbations in the logit space propagate across multiple autoregressive steps, affecting downstream tokens and overall output coherence, remain scarce.

Our work builds upon these insights by systematically injecting controlled perturbations into the logit space and empirically analyzing the resulting error propagation, robustness, and error-correction behavior in autoregressive LLMs, bridging the gap between adversarial attack research and communication-theoretic interpretations of language model behavior.

3 Setup

Throughout our study, we conduct small-scale experiments using the open-source GPT2-small model (124 million parameters). We develop a custom pipeline that performs greedy decoding, generating one token at a time. We deliberately disable top-k and top-p (nucleus) sampling in order to isolate and observe how injected errors propagate through the forward pass, without introducing the additional variability associated with stochastic sampling methods.

We evaluate the results against the ground truth text on the following metrics:

1. **BLEU Score**
   The Bilingual Evaluation Understudy (BLEU) score is a precision-based metric commonly used in machine translation and text generation tasks. It measures the degree of n-gram overlap between the generated output and the reference text. A higher BLEU score indicates greater lexical similarity, suggesting that the model's output closely matches the reference in terms of word choice and order. In our experiments, BLEU serves as a proxy for surface-level correctness.

2. **Perplexity Score**
   Perplexity quantifies how well a probability model predicts a sample. It is calculated as the exponentiation of the negative average log-likelihood of the true tokens under the model's predicted distribution. Lower perplexity values indicate that the model assigns higher probabilities to the correct next tokens, reflecting better fluency and internal consistency. We compute perplexity over the generated sequence to measure how confident the model is in its own predictions under perturbation.

3. **Semantic Similarity**
   To evaluate the meaning-level coherence of generated text, we measure semantic similarity between the output and the ground truth using sentence embeddings. Specifically, we use the Universal Sentence Encoder (USE) from TensorFlow Hub. The cosine similarity between the embedding vectors of the generated and reference sentences provides an estimate of how semantically close they are, regardless of exact wording. This metric captures cases where the generated text may differ lexically from the reference but still conveys the same meaning.

4 Experiments and Analysis

We experiment with various adversarial attack strategies on the model to evaluate its robustness. For each method, we plot the corresponding values of the evaluation metrics discussed earlier. To test our experimental hypotheses, we systematically vary key hyperparameters within the attacks and analyze how these changes influence the model's performance. We also provide intuitive explanations for the observed trends in the evaluation metrics.

4.1 Forcing Low Probablitiy tokens

4.1.1 Variant 1

In this experiment, we intentionally select a low-probability token at each step during token generation. After processing the context, we obtain the logits for the predicted token and sort them in descending order of probability. The chosen token corresponds to the specified index in this sorted list: an index of 1 represents greedy decoding (the most probable token), an index of 2 corresponds to the second most probable token, and so forth. We find that when the adversary forces selection of the 10th most probable token, both the BLEU score and semantic similarity significantly decline, while the perplexity dramatically increases to around 100 indicating a substantial loss of confidence in the generated output. We also parse these outputs into ChatGPT to rate them on a scale of 0-1 for hallucinated text (0 implying ground truth and 1 implying hallucinated).

Prompt

"The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is"

Ground Truth from Wikipedia

"The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is disagreement about its theoretical basis and the parameters required to produce the effect. The Mpemba effect is named after Tanzanian Erasto Bartholomeo Mpemba, who described it in 1963 as a secondary school student. The initial discovery and observations of the effect originate in ancient times; Aristotle said that it was common knowledge."

Output

Index 1: **Score = 0.1**
The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is a strong correlation between the temperature of a liquid and the amount of heat it can absorb. The Mpemba effect is also the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is a strong correlation between the temperature of a liquid and the amount of

Index 5: **Score = 0.6**
The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is evidence from laboratory observations, including that water can become a "coolant of heat." It may therefore not occur naturally when it has no liquid, because liquid freezes more readily and rapidly in water with water temperature at its lowest temperatures and in ice and solid rock environments with temperature and moisture that vary greatly in size or composition and temperature (3-8),

Index 10: **Score = 0.95**
The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is much controversy among theorists whether there's more such thermochemical phenomenon to heat in hot fluids; if we think thermologists believe so much is just fine when a gas that freezes rapidly turns black on one corner in heat waves like Katrina—but this may very plausually occur on all fronts! We need only remember some more important examples that suggest otherwise. . . the

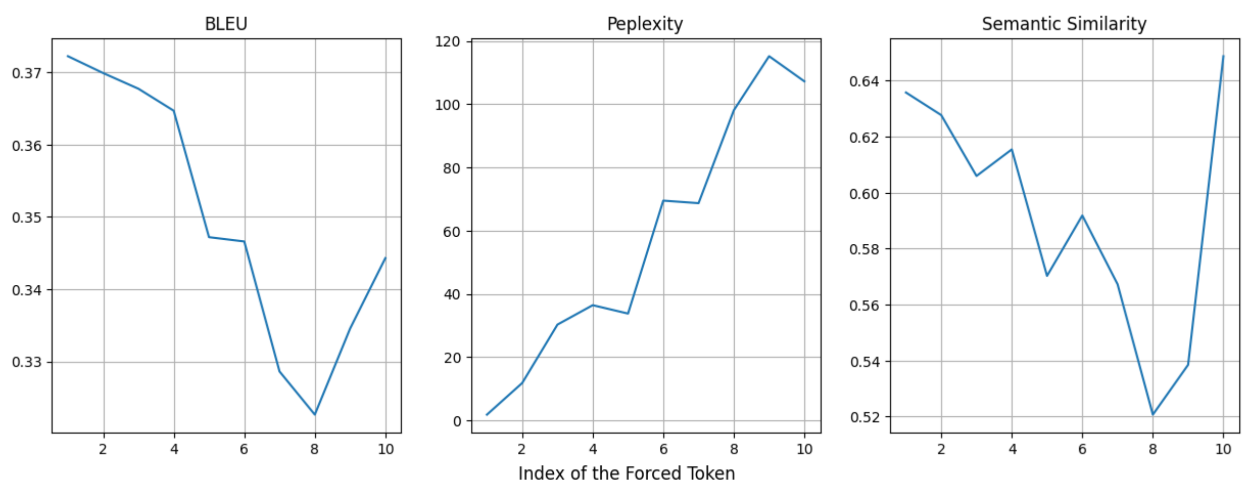

Figure 3: Score vs Index of Forced token

4.1.2 Variant 2

Following a similar framework to Variant 1, we again force low-probability tokens, but not at every token inference. Instead, we sample from a uniform proposal distribution to decide when to inject errors into the model, making the error source non-deterministic and random. As the acceptance probability for the proposal (alpha) increases, we observe a decline in both the BLEU score and semantic similarity, while the perplexity rises. However, the increase in perplexity is less pronounced than in Variant 1, reflecting the lower overall amount of injected errors.

Additionally, because errors are injected less frequently, the resulting degradation in output quality reflects a more realistic scenario where mistakes or uncertainties happen intermittently, rather than continuously. This helps differentiate the model's ability to recover from occasional errors versus handling persistent adversarial conditions, providing deeper insight into its error-correction capabilities.

We also performed additional trials by forcing the selection of the 20th and 30th most probable tokens to ensure the reproducibility of the results, and observed a consistent trend across all the scores.

Prompt

"The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is"

Output

Forcing 10th token with a probability of 0.5:
The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is a direct effect at that point as if the freezing of the liquid is instantaneous. For example, if liquid 1 is boiling at a speed comparable from 0 to 1.5 degrees Celsius per thousandth, then there has been little warming of air from the coldest part of the fluid to the hotest part. The temperature of liquids will be about the same as the temperature of water or other liquids if one assumes water-ice equilibrium is maintained. Thus in our laboratory (

Forcing 20th token with a probability of 0.5:
The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is a strong correlation between cooling and subsequent freezing which is why hot objects are more hot after heating cold matter. Heat causes the material to change its position by breaking contact lenses.
The Mnemonic A very general observation is often made that the Mpembi would need heat to freeze as rapidly without losing some momentum of ice when wet. This theory is quite hardline and based very much less highly on the reality of an ice globe than on anything else.

Forcing 30th token with a probability of 0.5:
The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is less of a difference: the lower the temperature, the further chilled the liquid. Water is more likely to freeze faster once it arrives at the base of the ice sheet, and the sooner can you melt the molten carbon which could be at the water that we are so happy eating. Also some minerals increase the melting rate on the higher liquid compared to the lower liquid ice meltrate resulting either from the mineralization within the sheet or simply oxidation of the other carbon due to hydro

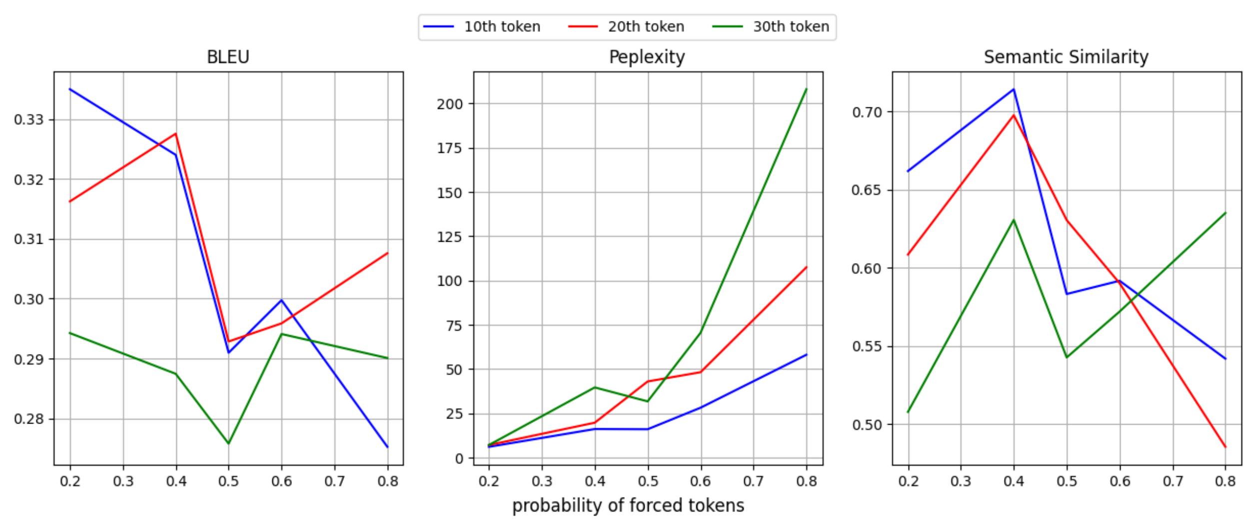

Figure 4: Forcing low probability tokens with a certain probability

4.2 Logit Perturbations

In this experiment we add noise sampled from different distributions to the logits generated from the model. At each token inference the noise sampled is independently from a pre-defined distribution. We perform a greedy decoding on the resulting perturbed logits and feed the generated token back into the context.

4.2.1 Addition of Gaussian Noise

During inference, we injected Gaussian noise with zero mean and varying variances into the logits generated by the model at each token prediction step followed by a greedy prediction. We observed that as the variance of the noise increased, both the BLEU score and semantic similarity exhibited an upward trend, while the perplexity also increased, indicating greater uncertainty in the model's predictions.

Possible reasons to this unexpected behavior could be the fact that adding noise to the logits can act as a form of regularization, encouraging the model to explore a wider range of token possibilities instead of being overly confident in its top predictions. This exploration may help the model avoid repetitive or overly deterministic outputs, thereby increasing diversity and potentially leading to generations that are more semantically aligned with reference texts—hence the rise in BLEU and semantic similarity scores.

However, the increase in perplexity suggests that the model's predictive distribution becomes more spread out and less certain, which is expected when noise is introduced. The model is less confident about its next token prediction, but this uncertainty might actually allow for better overall sentence-level coherence and fluency in certain contexts, particularly when the base model is otherwise too rigid.

Prompt

"The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is"

Output

Variance = 0.2
he Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is a strong correlation between the temperature of a liquid and the amount of water it freezes. The temperature of a liquid is the temperature at which it freezes. The temperature of a liquid is the temperature at which it freezes. The temperature of a liquid is the temperature at which it freezes. The temperature of a liquid is the temperature at which it freezes. The temperature of a liquid is the temperature at which it freezes. The temperature of a liquid is the temperature at which it freezes

Variance = 0.5
The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is a strong correlation between the temperature of the liquid and the amount of water it freezes. The Mpemba effect is also a consequence of the fact that the temperature of the liquid increases with the amount of water it freezes. The Mpemba effect is also a consequence of the fact that the temperature of the liquid increases with the amount of water it freezes.

Variance = 0.8
The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is no evidence of a strong Mpemba effect.
The Mpemba effect may be related to the fact that it is a property of water to freeze at the temperature of the liquid, rather than the other way around. The Mpemba effect can also be a result of the fact that the water is cold, rather than cold, as is the case with water. This is due to the fact that the water is not cold, but warm

4.2.2 Addition of Uniform Noise

We then experimented with adding uniform noise instead of Gaussian perturbations to the logits during inference. As the magnitude of the uniform noise increased, we observed a consistent decrease in BLEU score and semantic similarity, alongside an increase in perplexity. These results aligned with our expectations, in contrast to the trends observed with Gaussian noise.

Although both Gaussian and uniform noises can be zero-centered, they differ in distribution shape. Gaussian noise has most of its mass concentrated near zero and decays quickly, making small perturbations more likely. This can introduce subtle variability that helps the model explore alternative predictions without drastically distorting its confidence.

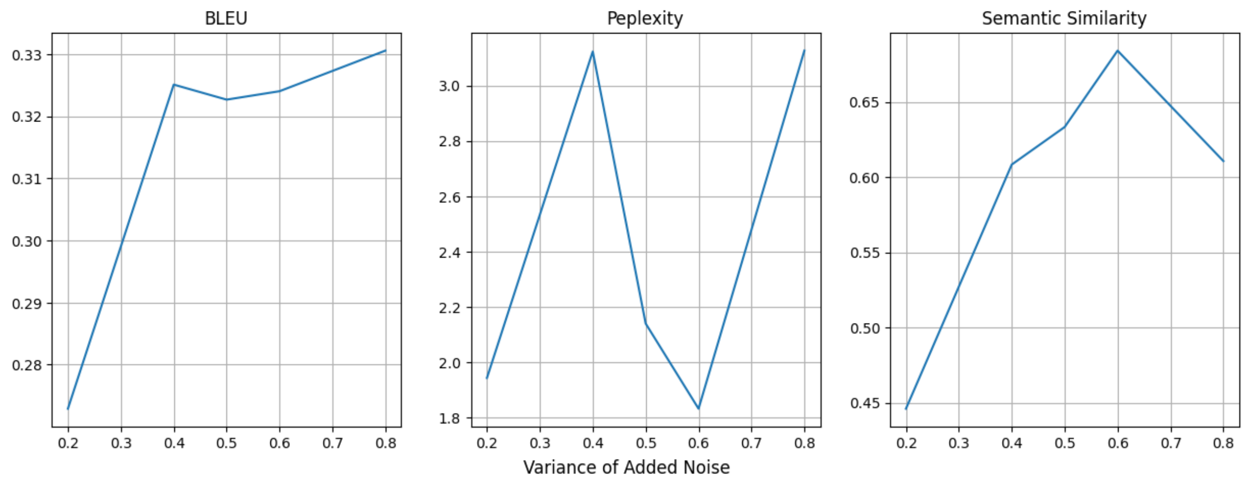

Figure 5: Scores against the variance of gaussian noise

In contrast, uniform noise distributes all values in the interval $U(-a, a)$ with equal probability. This means relatively large deviations from the original logits are just as likely as small ones, leading to more unpredictable changes in the model's output distribution. As a result, token predictions become less reliable, causing a decline in BLEU and semantic similarity, and an increase in perplexity due to heightened uncertainty.

Prompt

"The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is"

Output

Variance = 0.2
The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is a strong correlation between the temperature of a liquid and the amount of water that it freezes. The Mpemba effect is also the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions.
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Variance = 0.8
The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is a strong correlation between the temperature of the liquid and the amount of water that it freezes. The temperature of the liquid is the same as the temperature of the water.
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4.2.3 Addition of Gumbel Noise

We further experimented with adding Gumbel and Laplace noise to the logits during inference. In both cases, we observed that as the magnitude of the noise increased, the BLEU score and semantic similarity consistently decreased, while perplexity increased. These trends indicate a degradation in output quality and increased uncertainty in the model's predictions.

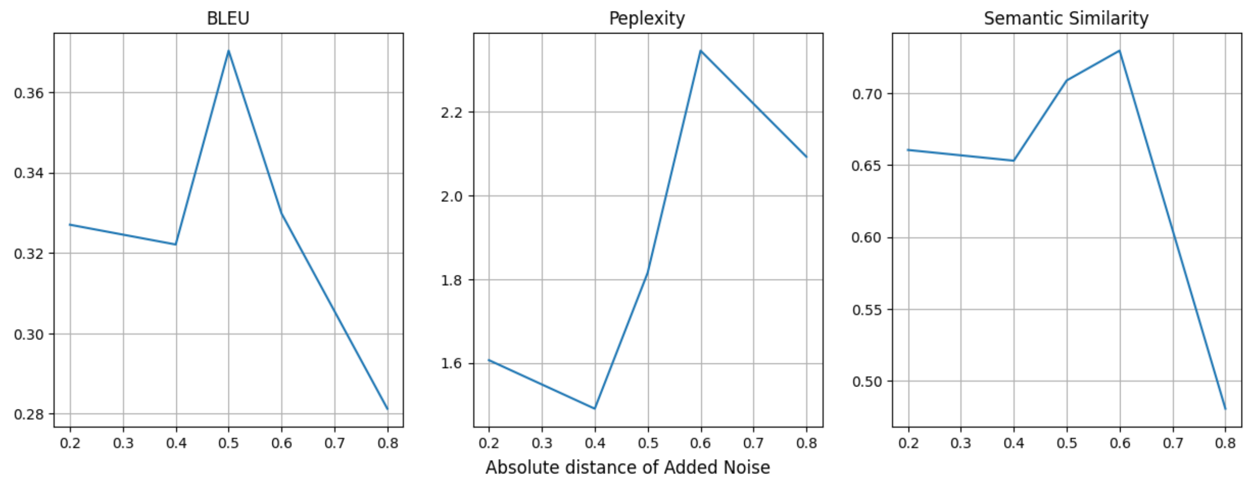

Figure 6: Score against the distance (a) of uniform noise $U(-a, a)$

4.3 Erasure Channel

In this experiment, we randomly remove certain tokens from the tokenized input prompt and feed the resulting incomplete prompt to the model. We then generate the model's output and compare it with the ground truth text. This setup evaluates the model's robustness to token erasures. We observe that, to some extent, the model successfully reconstructs the missing content, particularly preserving the overall language semantics. But we observe more repetitions which could be possibly because of the greedy decoding.

As we increase the probability of token erasures, the prediction accuracy declines. Correspondingly, the bleu score decreases, the perplexity score increases and semantic similarity decreases , aligning with the expected behavior of theoretical erasure channels.

Prompt

"The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is"

Ground Truth from Wikipedia

"The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the same liquid which begins cold, under otherwise similar conditions. There is disagreement about its theoretical basis and the parameters required to produce the effect. The Mpemba effect is named after Tanzanian Erasto Bartholomeo Mpemba, who described it in 1963 as a secondary school student. The initial discovery and observations of the effect originate in ancient times; Aristotle said that it was common knowledge."

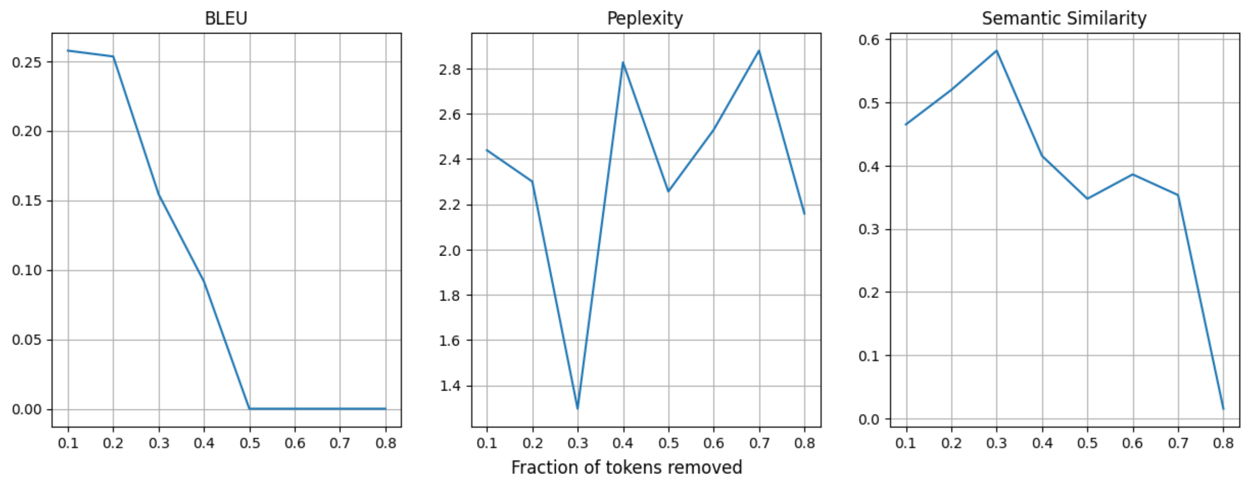

Figure 7: Scores VS Fraction of tokens erased

Prompt after Erasure

10% of tokens droppped:
"The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the liquid which begins cold under otherwise conditions. There is"

Output
"The Mpemba effect is the observation that a liquid (typically water) that is initially hot can freeze faster than the liquid which begins cold under otherwise conditions. There is a strong correlation between the temperature of the liquid and the amount of water in the liquid. The temperature of the liquid is the temperature at which the liquid freezes. The temperature of the liquid is the temperature at which the liquid freezes. The temperature of the liquid is the temperature at which the liquid freezes. The temperature of the liquid is the temperature at which"

20% of tokens droppped:
"he Mpemba effect is the observation that a (water) that is initially can than the same liquid which begins cold, under otherwise similar conditions There"

Output
he Mpemba effect is the observation that a (water) that is initially can than the same liquid which begins cold, under otherwise similar conditions There is a strong correlation between the temperature of the water and the amount of water in the water. The water is warmer in the Mpemba effect. The water is colder in the Mpemba effect. The water is warmer in the Mpemba effect. The water is warmer in the Mpemba effect. The water is

50% of tokens droppped:
Mpemba is the observation a liquid water) hot can faster the same liquid conditions.

Output
"Mpemba is the observation a liquid water) hot can faster the same liquid conditions. The water temperature is the temperature at which the water is heated. The temperature of the water is the temperature at which the water is cooled. The temperature of the water is the temperature at which the water is heated. The temperature of the water is the temperature at which the water is cooled. The temperature of the water is "

5 Conclusions and Future work

Our experiments demonstrate that adversarial attacks like forcing low probability tokens and injecting noise into model logits during inference has a significant impact on generation quality, with the nature of the noise distribution playing a critical role.

1. The model seems to be error tolerant especially in maintaining the linguistic semantics. While the perplexity and the semantic similarity to the ground truth text worsen with the probability of the forced tokens
2. Gaussian noise, despite increasing perplexity, unexpectedly led to improvements in BLEU score and semantic similarity, suggesting that small, centered perturbations can help the model explore more diverse yet semantically relevant outputs.
3. In contrast, uniform, Gumbel, and Laplace noise consistently caused BLEU scores and semantic similarity to decrease, while perplexity increased, indicating a decline in generation quality and greater uncertainty in predictions.
4. Erasure experiment showed that the model has some error correction capacity and produces text of similarity score 0.5 when the percentage of dropped tokens is 10% (which is generally the expected amount of erasures in channel coding theory).
5. We observe repeating sequences which is the virtue of the greedy decoding. Controlling experiments to a greedy framework helped reduce the stochasticity in model prediction and study only the effect of injected errors in the model's belief distribution. Future work would be extending these experiments to the top-k and top-p sampling systems.

Some questions [Abhi]:

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